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0.1 A note on background and scope

A first course on analysis that covers metric spaces is sufficient, familiarity with proofs is desirable. Level 7.4 provides this.

Note that you have likely heard that topology is about donuts and coffee mugs. Unfortunately, it's not as cool as that, in this course it is just about open sets and closed sets. The good news, though, is that in later courses such as Algebraic Topology, you will finally get the donuts and coffee mugs you were promised.

If you are reading this for the Cambridge course, the stuff that should be done in section §4.2 will be done in my algebraic topology notes instead, since it is only in that course that the triangulation stuff is done properly.

1 Basic properties of topological spaces

1.1 Definitions

A topological space is a generalization of a metric space. It is defined as follows.

First, denote by $P(X)$ the power set of X which is the set of all subsets of X.

Definition. Let X be a set. A topology on X is a set of subsets of X that we declare to be open. It is a subset $T \subseteq P(X)$ or $\tau \in P(X)$ or $\mathcal{T} \in P(X)$ satisfying

1. T contains the empty set

2. T contains X

3. The union of a collection of elements of T is in T

4. Finite intersections of collections of elements of T are in T

Notice that open sets in a metric space satisfied all these properties (I proved this in my Analysis notes), so this generalizes that. If X has a metric, it has a topology *induced* by that metric.

We can also generalize the notion of continuity. Recall that in metric spaces, a function is continuous if pre-images of open sets are open. We therefore define the following:

Definition. A function $f : X \to Y$ between topological spaces is called continuous if whenever V is an open subset of Y, its preimage $f^{-1}(V)$ is an open subset of X.

Definition. A function $f : X \to Y$ between topological spaces is said to be a homeomorphism if it is a bijection and it is continuous and its inverse is continuous. This is essentially an isomorphism between topological spaces.

Definition. A subset $S \subseteq X$ is said to be closed in X if $X \setminus S$ is open.

Definition. If $T \subseteq T'$ are topologies on a set X then we say that T is coarser than T' and that T' is finer than T *("finer topologies have more open sets")*

The statement that T' is finer than T is equivalent to the statement that the identity function from (X, T') to (X, T) is continuous.

Definition. Let (X, T) be a topological space and let $x_1, x_2, x_3, ...$ be a sequence in X. We say the sequence converges to $x_\infty \in X$ if for every open set U containing x_∞ (i.e. we also define this as an open neighbourhood of x_∞) there is some N such that $x_n \in U$ for all $n \geq N$.

Note that whether a sequence converges doesn't actually depend on the metric but only on the topology. Note that Lipschitz equivalent metrics have the same topology.

Example. An easy check shows that the function sending $(0, 1)$ to $\tan\left(\pi\left(x - \frac{1}{2}\right)\right)$ is continuous and has a continuous inverse (with the standard metric topologies) and it therefore defined a homeomorphism $(0, 1) \to \mathbb{R}$ so these spaces are, in a sense, topologically equivalent, but their metrics are certainly not equivalent. Also, completeness is not a topological property because $(0, 1)$ is not complete but $\mathbb{R}$ is complete. Completeness depends on the underlying metric.

Another way to see that completeness is not a topological property is to note that there is no natural way to define a Cauchy sequence on a general topological space without additional structure.

Example. The discrete topology on a set X, T_{disc}, is the topology induced by the metric $d(x, y) = 1 - \delta_{xy}$, in other words 1 unless the points are the same. In this topology, every subset is open. This is the finest topology we can induce on a set X.

Example. The indiscrete topology on a set X is the topology consisting only of X and the empty set. It is the coarsest topology we can induce on a set X. Note that it cannot be induced by a metric if X has more than 2 points, since if $x, y \in X$ and we suppose T_{indisc} is induced by a metric d, then set $d(x, y) = \varepsilon$, then the open ball $B_{\frac{\varepsilon}{2}}(x)$ is open and contains x but not y, so this metric topology is not the indiscrete topology.

Also, in the indiscrete topology, every sequence converges to every point, by definition, since X itself is the only open neighbourhood about any point.

Example. Let X be a set, then the collection $T_{\text{cofinite}} = \{X \setminus F : F \subseteq X \text{ is finite}\} \cup \{\emptyset\}$, which is called the cofinite topology, is a topology. This is a topology because finite intersections and arbitrary unions of elements in T_{cofinite} are in T_{cofinite} and so is X and the empty set.

Ok, so now that we have the basic definitions down, I have to show this.

Theorem. There are infinitely many prime numbers (This theorem is well known from earlier levels, but we can prove it using topological spaces as follows)

Proof. Define a topology on the integers $\mathbb{Z}$ by declaring a subset $U \subseteq \mathbb{Z}$ to be open if and only if it is a union of arithmetic sequences. Note that the empty set counts, as it can be seen as a nullary union. Specifically, for $a \neq 0$ define $S(a,b) = \{an + b | n \in \mathbb{Z}\} = a\mathbb{Z} + b$ and make U be open if and only if it is a union of these sets. Equivalently, U is open if and only if for all x there is some non-zero integer a such that $S(a,x) \subseteq U$. To check that this is a topology, note that

- The empty set is open by definition and $\mathbb{Z}$ is open as it is $S(1,0)$.

- Any union of unions of arithmetic sequences is a union of arithmetic sequences

- The intersection of two (and hence finitely many) open sets is open: Let U_1 and U_2 be open sets and let $x \in U_1 \cap U_2$ (with numbers a_1, a_2 establishing membership as described above), then $a = \operatorname{lcm}(a_1, a_2)$ has $S(a,x) \subseteq S(a_i, x) \subseteq U_i$ for $i = 1, 2$.

This topology has the property that a finite non-empty set cannot be open, since arithmetic sequences are infinite. This implies that the compliment of a finite non-empty set cannot be a closed set.

This topology also has the property that the basis sets $S(a,b)$ are both open and closed: To see that they are closed, note that $S(a,b) = \mathbb{Z} \setminus \bigcup_{j=1}^{a-1} S(a, b+j)$. The only integers that are not integer multiplies of prime numbers are -1 and +1 i.e. $\mathbb{Z} \setminus \{-1, +1\} = \bigcup_{p \text{ prime}} S(p, 0)$. Now by earlier discussion, the left hand side cannot be closed. On the other hand, each set $S(p,0)$ is closed. If there were only finitely many prime numbers, the right hand side would be a finite union of closed sets which is closed. This would be a contradiction so there are infinitely many prime numbers.

□

Example. Let $X = \{o, c\}$ and let $T := \{\emptyset, \{o\}, \{o, c\}\}$. This is the Sierpinski space.

Example. On $\mathbb{R}$, we can define the right order topology as $T_{\text{ord}} := \{(a, \infty) : a \in [-\infty, \infty]\}$. The axioms for a topology are verified as follows. If $\{(a_\alpha, \infty)\} : \alpha \in I$ is a collection of open sets then:

$$\bigcup_{\alpha \in I} (a_\alpha, \infty) = \left(\inf_{\alpha \in I} a_\alpha, \infty \right)$$

If I is finite:

$$\bigcap_{\alpha \in I} (a_\alpha, \infty) = \left(\max_{\alpha \in I} a_\alpha, \infty \right)$$

Example. Consider the cocountable topology on $\mathbb{R}$ which is defined as

$$T_{\text{cocountable}} = \{X \setminus F : F \subseteq X \text{ is countable}\} \cup \{\emptyset\}$$

This is topology with the same proof as in the cofinite space and the closed subsets are precisely $\mathbb{R}$ and the countable sets. In this topology, if x_n is a sequence and $x \in \mathbb{R}$ is a point then $\{x\} \cup (\mathbb{R} \setminus \{x_n\})$ is an open neighbourhood of x. This contains all but finitely many elements of the sequence if and only if the sequence is eventually constant x. Therefore, the only convergent sequences in $\mathbb{R}$ in the cocountable topology are the eventually constant sequences.

1.2 Basis and subbasis

Definition. Let T be a topology on X. Then a basis for T is a subset $B \subseteq T$ such that every element of T is a union of elements of B.

Note that from this definition it follows that a basis must cover the space.

Definition. A subbasis for T is a subset $S \subseteq T$ such that every element of T is obtained by taking a union of sets obtained as finite intersections of elements of S. Equivalently, T is the coarsest topology containing S .

Example. In any metric topology the set of open balls is both a basis and a subbasis

If S is any collection of subsets of X which cover X then there is a unique topology T on X for which S is a subbasis as shown in the following proposition

Proposition. The collection of unions of finite intersections of elements of S is a topology.

Proof. It contains the empty set as a nullary union. It contains X because S covers X. It is closed under unions because "unions of unions of sets with property X" fall under "unions of sets with property X". For finite intersections it suffices to prove that for any $A, B \in \mathcal{T}$, the intersection $A \cap B \in \mathcal{T}$. Let $A = \bigcup_{i \in I} \bigcap_{j=1}^{n_i} S_{i,j}$ and $B = \bigcup_{k \in K} \bigcap_{l=1}^{m_k} S'_{k,l}$, where $S_{i,j}, S'_{k,l} \in \mathcal{S}$. We have

$$A \cap B = \left(\bigcup_{i \in I} \bigcap_{j=1}^{n_i} S_{i,j} \right) \cap \left(\bigcup_{k \in K} \bigcap_{l=1}^{m_k} S'_{k,l} \right) = \bigcup_{i \in I, k \in K} \left(\bigcap_{j=1}^{n_i} S_{i,j} \cap \bigcap_{l=1}^{m_k} S'_{k,l} \right)$$

The resulting expression is explicitly a union of finite intersections of elements of $\mathcal{S}$. Therefore, $A \cap B \in \mathcal{T}$.

□

Proposition. Let $f : (X, T_X) \to (Y, T_Y)$ be any function and let $S \subseteq T_Y$ be a subbasis. If $f^{-1}(U) \in T_X$ for each $U \in S$ then f is continuous.

Proof. If V is open in Y then by definition of a subbasis we have $V = \bigcup_{a \in I} V_a$ for a set I where V_a is $\bigcap_{b \in J_a} U_{a,b}$ for finite sets J_a and subbasis elements $U_{a,b} \in S$. Then we have

$$f^{-1}(V) = \bigcup_{a \in I} f^{-1}(V_a) = \bigcup_{a \in I} \left(\bigcap_{b \in J_a} f^{-1}(U_{a,b}) \right)$$

This is open because each $\bigcap_{b \in J_a} f^{-1}(U_{a,b})$ is open by assumption and the intersections are finite.

□

Example. Note that the set of open cubes $(a_1, b_1) \times (a_2, b_2) \times ... \times (a_n, b_n)$ is a basis for the standard metric topology on $\mathbb{R}^n$. The reason is because in an open set there is an open cube about each point and open cubes are open so the open sets are exactly the unions of open cubes. Also, since finite intersections of open cubes are open cubes, it is also a subbasis.

Definition. A topology is said to be second countable if it has a countable basis.

Example. The metric topology on $\mathbb{R}^n$ is second countable. To see this, it is an exercise to check that all open subsets of $\mathbb{R}^n$ are unions of open balls with rational center and radius.

1.3 Closure and interior

Note that intersections and finite unions of closed sets are closed, because the reverse properties hold for open sets. Also, $\emptyset$ and X are always closed.

Proposition. A function between topological spaces is continuous if and only if pre-images of closed sets are closed.

Proof. Suppose that for all C closed, $f^{-1}(C)$ is closed. Then for all $X \setminus C$ open, $f^{-1}(X \setminus C) = f^{-1}(X) \setminus f^{-1}(C)$ is open. The converse is proved the same way.

□

Since intersections of closed sets are closed, given a set A, the intersection of all closed sets containing A is closed. This is called the closure of A and we denote it $\bar{A}$.

Example. The closure of $(0, 1)$ in $\mathbb{R}$ is $[0, 1]$.

The interior A° of a set A is the largest open set of X contained in A. This exists as the union of all open sets contained in A. Also, the interior of A is the complement of the closure of $X \setminus A$.

Example. The interior of $[0,1]$ in $\mathbb{R}$ is $(0,1)$.

Definition. Let X be a topological space and let $A \subseteq X$. Then a point $x \in X$ is a limit point of A if there is a sequence x_n in A such that $x_n \neq x$ for all n and $x_n \to x$.

Note that we have seen (for example with the discrete topology) that limits need not be unique. However, they are unique if the space is Hausdorff - we will discuss that in §1.4.

Proposition. If $C \subseteq X$ is a closed subset of a topological space then its limit points are in C.

Proof. Let x_n be a sequence in C converging to $x \in X$. If $x \neq C$ then $x \in X \setminus C$ is an open neighbourhood of x so by definition of convergence, $x_n \in X \setminus C$ for large enough n which is a contradiction.

□

In particular, every limit point of a subset is in its closure.

Definition. Let X be a topological space and A be a subset. Then A is said to be dense in X if $\bar{A} = X$.

Example. $\mathbb{Q}$ is dense in $\mathbb{R}$.

Proposition. (Alternative definition of dense) A subset of a topological space is dense if and only if its intersection with every non-empty open set is non-empty.

Proof. Suppose a subset A is dense but for some open set U, $A \cap U = \emptyset$. Then $X \setminus U$ is a closed subset of X containing A that is not X, so the closure of A is not X, contradicting denseness.

Conversely, suppose a set A has the property that its intersection with every non-empty open set is non-empty. Then any closed set containing A is the complement of an open set not containing any element of A, but the only such open set is the empty set.

□

1.4 Hausdorff spaces

Definition. A topological space (X, T) is said to be Hausdorff if for each pair $x, y \in X$ of distinct points there are disjoint open neighbourhoods around x and y.

This basically means x, y can be separated. Note that a topological space arising from a metric space is Hausdorff, because if x, y are distinct then the open balls about them of half their distance are the open sets we want.

Proposition. Limits are unique in Hausdorff topological spaces.

Proof. Suppose that x, x' are both limits of $\{x_n\}$. Then there are disjoint open neighbourhoods $U \ni x$ and $V \ni x'$. Since $x_n \to x$ only finitely many x_n's are outside U but also only finitely many x_n's are outside V so the sequence cannot converge to both x and x'.

□

Proposition. In a Hausdorff topological space the one-point sets are closed

Proof. Let X be a Hausdorff space and $x \in X$. For each $y \neq x$ we can find disjoint open neighbourhoods $U_y \ni y$ and $V_y \ni x$ and in particular $X \setminus U_y$ is closed. The intersection over all $y \neq x$ of $X \setminus U_y$ is just x and the result follows as intersections of closed sets are closed.

□

Example. The cofinite topology on an infinite set is not Hausdorff because an open neighbourhood of $x \in X$ is a set of the form $X \setminus F$ with F finite and another disjoint open neighbourhood about any point must be in F which is not possible.

1.5 New topologies from old

1.5.1 Subspace topology

Definition. Let X be a topological space and let $Y \subseteq X$ be a subset. The subspace topology on Y is the set $\{U \cap Y : U \text{ is open in } X\}$.

Note that this is a topology because $\bigcup_{a \in I} (U_a \cap Y) = \left(\bigcup_{a \in I} U_a\right) \cap Y$ and similarly for finite intersections.

We write the subspace topology as $T_{X|Y}$.

Definition. The inclusion $i : \left(Y, T_{X|Y}\right) \to (X, T_X)$ is continuous. Also, the subspace topology is the coarsest topology on Y for which i is continuous.

Proof. If U is open in (X, T_X) then $i^{-1}(U) = U \cap Y$ is open in $\left(Y, T_{X|Y}\right)$ by definition so i is indeed continuous.

If T' is a topology on Y such that $i : (Y, T') \to (X, T_X)$ is continuous then for each $U \in T_X$ we have $i^{-1}(U) = U \cap Y \in T'$ so $T_{X|Y} \subseteq T'$ so the subspace topology is the coarsest topology on Y for which i is continuous.

□

Note that for any topological space Z, if $Y \subseteq X$ then a function $f : Z \to Y$ is continuous if and only if the function $g : Z \to X$ defined by sending each element of Z to $f(Z) \in Y \subseteq X$ is continuous. In other words, to give a continuous function $f : Z \to Y$ is the same as to give a continuous function $f : Z \to X$ whose image happens to land in Y.

Note that if U is open in X and a set V is open in U with the subspace topology then V is open in X. This follows from the fact that $V = W \cap U$ for some open W by definition of the subspace topology, hence open.

Theorem. (Gluing lemma) Let $f : X \to Y$ be a function between topological spaces. Then

1. If there is a cover $\{U_a : a \in I\}$ of X by open sets and the restrictions $f|_{U_a} : U_a \to Y$ are continuous for each $a \in I$ then f is continuous.

2. If there is a finite cover $\{C_a : a \in I\}$ of X by closed sets and the restrictions $f|_{C_a} : C_a \to Y$ are continuous for each $a \in I$ then f is continuous.

Proof. Let $V \subseteq Y$ be open. As the U_a make a cover we have

$$f^{-1}(V) = \bigcup_{a \in I} \left(f^{-1}(V) \cap U_a\right) = \bigcup_{a \in I} \left(f|_{U_a}\right)^{-1}(V)$$

Since each $f|_{U_a}$ is continuous we have $\left(f|_{U_a}\right)^{-1}(V)$ is open so $f^{-1}(V)$ is open as a union of open sets so f is continuous.

The closed set version is analogous noting that finite unions of closed sets are closed.

□

Definition. Define S^n as $\left\{x \in \mathbb{R}^{n+1} : \|x\| = 1\right\}$. This is the n-dimensional sphere and it is given the subspace topology from the standard topology on $\mathbb{R}^{n+1}$, or the induced metric topology. In particular, S^1 is the circle.

Definition. D^n is the closed unit disc and it is given by $\{x \in \mathbb{R}^n : \|x\| \leq 1\}$, again with the subspace topology, and this contains S^{n-1} as a subspace.

1.5.2 Quotient topology

Let (X, T_X) be a topological space and let $\sim$ be an equivalence relation on X with the set of equivalence classes $X_{/\sim}$ and the quotient map $\pi : X \to X_{/\sim}$. Then define

$$T_{X/\sim} := \left\{U \subseteq X_{/\sim} : \pi^{-1}(U) \in T_X\right\}$$

Proposition. This is actually a topology

Proof. It is an exercise to check that the empty set and the whole space are in $T_{X/\sim}$. If U_a is a collection of elements of $T_{X/\sim}$ then

$$\pi^{-1}\left(\bigcup_{a\in I} U_a\right) = \bigcup_{a\in I} \pi^{-1}(U_a) \in T_X$$

so $\bigcup_{a\in I} U_a$ is in $T_{X/\sim}$. The argument for finite intersections is analogous.

□

Proposition. The quotient map $\pi : (X, T_X) \to \left(X_{/\sim}, T_{X/\sim}\right)$ is continuous. In fact, $T_{X/\sim}$ is the finest topology on $X_{/\sim}$ for which π is continuous.

Proof. Immediate from the definition

□

An alternative characterization of the quotient topology is that for any topological space (Z, T_Z) a function $f : \left(X_{/\sim}, T_{X/\sim}\right) \to (Z, T_Z)$ is continuous if and only if the composition

$$(X, T_X) \xrightarrow{\pi} \left(X_{/\sim}, T_{X/\sim}\right) \xrightarrow{f} (Z, T_Z)$$

is continuous.

In other words, to give a continuous function f out of the quotient space $X_{/\sim}$ to a space Z is the same as giving a continuous function $g : X \to Z$ such that $g(x) = g(x')$ whenever $x \sim x'$. In other words, this says that $g = f \circ \pi$ for a unique function $f : X_{/\sim} \to Z$.

Definition. A continuous map $f : X \to Y$ is a quotient map if it is surjective and $U \in T_Y \iff f^{-1}(U) \in T_X$.

If $f : X \to Y$ is a quotient map and $\sim$ is the equivalence relation on X given by $x \sim x' \iff f(x) = f(x')$ then the above discussion gives a homeomorphism $f : X_{/\sim} \to Y$.

Example. Consider the equivalence relation on $\mathbb{R}$ given by $x \sim y \iff x - y \in \mathbb{Z}$. The function $f : \mathbb{R} \to \mathbb{R}^2$ sending $x \mapsto (\sin(2\pi x), \cos(2\pi x))$ is continuous and has image in S^1 so we may consider it as a continuous function into S^1. Also, $x \sim y$ exactly when $f(x) = f(y)$, and in fact f defines a homeomorphism $\mathbb{R}_{/\sim} \to S^1$.

Example. Consider the subspace $X = \mathbb{R} \times \{0, 1\}$ as a subspace of $\mathbb{R}^2$ and the equivalence defined by $(x, i) \sim (y, j)$ whenever they are equal or $x = y \neq 0$. Then this is a "line with two origins": It looks like figure 1

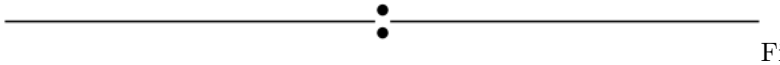


Figure 1

The points $[(0, 0)]$ and $[(0, 1)]$ are distinct. An open neighbourhood is a set such that $\pi^{-1}(U)$ is open in $\mathbb{R} \times \{0, 1\}$. This means that any neighbourhood of $(0, 0)$ must contain $(-\varepsilon, \varepsilon) \times \{0\}$ for some ε, but it also must be a union of equivalence classes so it must also contain $((-\varepsilon, \varepsilon) \setminus (0, 0)) \times \{1\}$. In particular, this space is not Hausdorff because the 2 origins cannot be separated, so this cannot be realized as a metric space.

Example. In $\mathbb{R}^n \setminus 0$, let $x \sim y$ if $x = \lambda y$ with $\lambda > 0$, then the quotient space is homeomorphic to the sphere S^{n-1}.

1.5.3 Product topology and box topology

Given sets X and Y we can define the projection formulas as $\pi_X : X \times Y \to X$ by sending $(x, y) \mapsto x$ and $\pi_Y : X \times Y \to Y$ by sending $(x, y) \mapsto y$.

Definition. We define the product topology as the coarsest topology that makes all the projection maps continuous. This means it is generated by the sets that are the whole space in all but one coordinate and an open set in one coordinate. This generalizes to products of infinite families of sets.

Definition. We define the box topology on $T_{X\times Y}$ as the topology generated by the product of an open subset of X and Y respectively.

It turns out that for finite products of sets, these definitions are equivalent.

Proposition. For products of finitely many topological spaces it agrees with the product topology.

Proof. For both of these directions we need to check inclusion of generating sets

Product topology contained in box topology: For a generating set of the product topology, it is an open set in one space times the whole of the rest of them, which is in the generating set of the box topology.

Box topology contained in product topology: For a product of open sets, since the product is finite, it is a finite intersection of a bunch of generating sets in the product topology.

□

It turns out that the subbasis for the box topology is also a basis. This fact follows from the following lemma:

Lemma. If a subbasis S for a topological space is closed under finite intersections then it is a basis

This is immediate from the definition of subbasis and basis.

In this case the "universal property" is: To give a continuous function from Z into $X \times Y$ is the same as to give a continuous function $Z \to X$ and $Z \to Y$. A continuous function into the product space is continuous when restricted to X as we compose with the continuous projection map. If Z is continuous into X, Y then pre-images of open subsets of X, which after pulling back to $X \times Y$ become pre-images of the generating open sets, are open, and similarly for Y, so the function $Z \to X \times Y$ is also continuous.

The product topology has the nice property that a sequence converges if and only if it converges in every coordinate. To prove this we use the following lemma.

Lemma. Continuous functions preserve sequence convergence

Proof. Let $x_n \to x$ in X and $f : X \to Y$ be continuous

Let V be an arbitrary neighbourhood of $f(x)$ in Y, then by convergence every x_n is eventually contained in $f^{-1}(V)$ as this is open by continuity, so every $f(x_n)$ is eventually in V.

□

Now here is how we prove that in the product topology a sequence converges if and only if it converges in every coordinate.

Each projection map $p_i : \prod X_j \to X_i$ is continuous by the definition of the product topology. Because continuous functions preserve sequence convergence, if a sequence (x_n) converges to x in the product space, then its coordinate projection $p_i(x_n)$ must converge to $p_i(x)$ in each factor space X_i.

Conversely, if a sequence converges in every coordinate, a basic open set in the product topology is a finite intersection of

sets of the form $p_i^{-1}(U_i)$, where U_i is an open set in X_i containing $p_i(x)$. If $p_i(x_n) \to p_i(x)$ for every coordinate i, then for any such neighbourhood, the sequence (x_n) eventually enters and stays inside $p_i^{-1}(U_i)$ for all n large enough, and thus inside the finite intersection for some n large enough. Thus the sequence converges in the product topology.

1.6 Continuous functions into $\mathbb{R}$

The following properties generalize quite nicely from metric spaces.

Proposition. Let X be a topological space and let $f, g : X \to \mathbb{R}$ be continuous functions. Then the functiosn $f + g$ and $f * g$ are also continuous.

Proof. $f + g$ can be realized as the composition $X \xrightarrow{(f,g)} \mathbb{R}^2 \xrightarrow{\text{add the coordinates}} \mathbb{R}$ which is a composition of continuous coordinates, hence continuous. The proof for $f * g$ is similar and left as an exercise.

□

Proposition. Let X be a topological space and let $f_n : X \to \mathbb{R}$ be a sequence of continuous functions converging uniformly to a function $f : X \to \mathbb{R}$. Then f is continuous.

Proof. Fix $x \in X$ and let $\varepsilon > 0$. By uniform convergence, there exists $N \in \mathbb{N}$ such that, for every $y \in X$,

$$|f_N(y) - f(y)| < \frac{\varepsilon}{3}.$$

Since f_N is continuous at x, there exists an open neighbourhood U of x such that, for every $y \in U$,

$$|f_N(y) - f_N(x)| < \frac{\varepsilon}{3}.$$

Therefore, for every $y \in U$, the triangle inequality gives

$$|f(y) - f(x)| \leq |f(y) - f_N(y)| + |f_N(y) - f_N(x)| + |f_N(x) - f(x)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$$

Thus f is continuous at x. Since $x \in X$ was arbitrary, f is continuous on X.

□

Note that these results hold in $\mathbb{C}$ with the same proof.

2 Connectedness and components

2.1 Conncetedness

Definition. A topological space is said to be disconnected if it can be written as $X = U \cup V$ where U and V are disjoint non-empty open subsets. It is connected if it is not disconnected.

Example. A space with the indiscrete topology is connected and a space with more than one point with the discrete topology is disconnected.

Example. The space $\mathbb{R} \setminus 0$ is disconnected because $(0, \infty)$ and $(-\infty, 0)$ are the open subsets.

Proposition. A space X is disconnected if and only if there exists a surjective continuous function $f : X \to (\{0, 1\}, T_{\text{disc}})$.

Proof. If X is disconnected with $X = U \cup V$ a decomposition into disjoint non-empty open subsets define the function $f(x) = 1_V$ i.e. f is 0 on U and 1 on V. Then the pre-images of the open subsets of $\{0, 1\}$ are $\emptyset, U, V, U \cup V$ which are all open so f is continuous and it is surjective as U, V are non-empty.

Conversely, if such an f exists then $U = f^{-1}(0)$ and $V = f^{-1}(1)$ gives a decomposition of X into disjoint non-empty open subsets.

□

Proposition. The spaces $[0,1], [0,1), (0,1)$ are connected.

Proof. If it were disconnected there would be a continuous surjection $f : [0,1] \to \{0,1\}$ contradicting the intermediate value theorem. Similar for $[0,1)$ and $(0,1)$.

□

Theorem. (Intermediate Value Theorem, topological version) Let X be connected and let $f : X \to \mathbb{R}$ be continuous. Then if there are points $x_0, x_1 \in X$ such that $f(x_0) < 0 < f(x_1)$ then there is an $x \in X$ with $f(x) = 0$.

Proof. If there was no such x then $f^{-1}(-\infty, 0)$ and $f^{-1}(0, \infty)$ would give 2 non-empty disjoint open subsets that cover X

□

In fact, since disconnected sets fail the IVT condition, so the above theorem can be stated as "X connected $\iff$ X satisfies IVT"

Note that continuous functions map connected sets to connected sets as shown by the following proposition

Proposition. If $f : X \to Y$ is a continuous surjection and X is connected then Y is connected

Proof. We will prove that if Y is disconnected then X is disconnected. Since Y is disconnected there is a continuous surjection $g : Y \to \{0,1\}$ which becomes a continuous surjection $g \circ f : X \to \{0,1\}$ so X is disconnected.

□

From this it follows that images of connected sets under continuous functions are connected.

Proposition. If $f : X \to Y$ is a homeomorphism and $Z \subseteq X$ is a subspace then $f|_Z : Z \to f(Z)$ is a homeomorphism.

Proof. This is continuous and maps open sets to open sets by definition of the subspace topology and f being a homeomorphism thus it is also a homeomorphism.

□

Example. The spaces $[0,1]$ and $(0,1)$ are not homeomorphic, because if they were homeomorphic then removing 0 gives a homeomorpsim $(0,1] \to (0,1) \setminus f(0)$ which goes from a connected thing to a disconnected thing which is not possible.

Example. $\mathbb{R}$ is not homeomorphic to $\mathbb{R}^2$ as only $\mathbb{R}^2$ stays connected after we remove a point - we will justify this in §2.2 on path-connectedness.

Proposition. Let $\{X_\alpha \subseteq X : \alpha \in I\}$ be a collection of subspaces of X such that each X_α is connected and $\bigcap_{\alpha \in I} X_\alpha$ is non-empty, then $\bigcup_{\alpha \in I} X_\alpha$ is connected.

Proof. Suppose $X = \bigcup_{\alpha \in I} X_\alpha$ and that $X = U \cup V$ is a decomposition into disjoint open subsets, then we want to show that one of U, V is empty. Each X_α is connected so one of $X_\alpha \cap U$ and $X_\alpha \cap V$ must be empty for each α, but all X_α's have a point in common so they lie either all in U or all in V so one of U, V is empty.

□

Corollary. If X, Y are connected then $X \times Y$ is connected.

Proof. We can do this by applying the above proposition twice. If either X, Y is empty so is the product which makes this trivial so assume both are non-empty. For $x \in X$ consider the sets $C_y := X \times \{y\} \cup \{x\} \times Y$. Then these are each connected by the proposition and the union of all of them, which is $X \times Y$, are also connected by the above proposition as they have $\{x\} \times Y$ in common.

□

2.2 Path-Connectedness

We can define a different notion of connectedness as follows.

Definition. If X is a topological space and $x_0, x_1 \in X$ then a path from x_0 to x_1 is a continuous function $\gamma : [0,1] \to X$ such that $\gamma(0) = x_0$ and $\gamma(1) = x_1$.

Definition. A topological space X is path-connected if for each $x_0, x_1 \in X$ there is a path from x_0 to x_1.

Example. $[0,1]$ and $(0,1)$ and $[0,1)$ are path-connected. The proof is left as an exercise.

Example. Any convex subset of $\mathbb{R}^n$ is path-connected because convex sets are defined such that each pair of points has not only a path but a straight path between them.

Example. $\mathbb{R}^2 \setminus 0$ is path-connected because between any 2 points we can do a straight line and if this crosses the origin do a semicircle instead. We will see next that path connected implies connected, and this will justify our earlier claim that $\mathbb{R}$ and $\mathbb{R}^2$ are not homeomorphic.

Theorem. If X is path-connected then it is connected.

Proof. Let X be path-connected and suppose for contradiction that $f : X \to \{0,1\}$ is a continuous surjective function and pick x_0, x_1 such that $f(x_0) = 0$ and $f(x_1) = 1$ and let $\gamma : [0,1] \to X$ be a path from x_0 to x_1. Then $f \circ \gamma : [0,1] \to \{0,1\}$ is continuous and surjective which is nonsense as we know continuous images are connected.

□

In §2.3 we will see an example of a space that is connected but not path-connected.

2.3 Components

If a space is not path-connected then it can be split into path-connected components: Define a relation $\sim$ such that $x \sim y$ whenever they are connected by a path. This is reflexive by the constant path and symmetric by the path that goes the other way, and to check that it is transitive, given a path γ from $x \to y$ and a path $\gamma' : y \to z$ define the path $\gamma + \gamma'$ by

$$(\gamma + \gamma')(t) := \begin{cases} \gamma(2t) & 0 \le t \le \frac{1}{2} \\ \gamma'(2t-1) & \frac{1}{2} \le t \le 1 \end{cases}$$

from x to z. This is continuous by the gluing lemma.

We can now define the path components as the equivalence classes of $\sim$.

Proposition. If we have a path component $C \subseteq X$ then C is actually path-connected itself i.e. there is a path between 2 points in C that lies in C itself.

Proof. If $\gamma : [0,1] \to X$ is a path from x_0 to x_1 in X then for each $t \in [0,1]$ the restriction $\gamma_t : [0,t] \cong [0,1] \to X$ is a path in X from x_0 to $\gamma(t)$ which therefore lies in the path component C as well, so the whole path takes place in the subspace C.

□

We would like to do the same but with "path-connected" replaced by "connected.

Define an equivalence relation $\approx$ on X by $x \approx y$ whenever there is a connected subspace $C \subseteq X$ containing both x and y.

Proposition. $\approx$ is an equivalence relation and its equivalence classes (the "connected components") are connected.

Proof. Reflexivity and symmetry are obvious from the definition. For transitivity, suppose that $x \approx y$ and $y \approx z$ and that C_1 is a connected space containing x, y and C_2 is a connected space containing y, z. Then their intersection contains y so is non-empty and hence $C_1 \cup C_2$ is connected by the proposition at the end of §2.1.

To see that the equivalence classes are connected, let $C \subseteq X$ be any equivalence class with respect to $\approx$ and suppose that $f : C \to \{0,1\}$ is a continuous surjection and pick x, y such that $f(x) = 0$ and $f(y) = 1$. Because $x \approx y$ there is a connected subspace D such that $x, y \in D \subseteq X$ but then each $d \in D$ has $d \approx x$ too so $D \subseteq C$, but D is connected and f restricted to D is still a continuous surjection into $\{0,1\}$ which is a contradiction.

□

Example. The space $\mathbb{R} \setminus 0$ has 2 connected components: the positive and negative reals.

Example. The subspace $\mathbb{Q} \subseteq \mathbb{R}$ has connected components equal to the one-point sets $\{x\}$.

By the above example, we see that the components are not necessarily open in the underlying space so be careful.

Example. (Topologists sine curve) Let $S := \left\{\left(x, \sin\left(\frac{1}{x}\right)\right) \in \mathbb{R}^2 : 0 < x \leq 1\right\}$

This is path-connected as the continuous image of a path-connected set. We will prove below that the closure of a connected subspace is connected, so $\overline{S}$ is connected, but as it turns out, $\overline{S}$ is connected but not path-connected and we will show this now. Note that $\bar{S} = S \cup (\{0\} \times [-1,1])$.

Obligatory image to visualize S: See figure 2

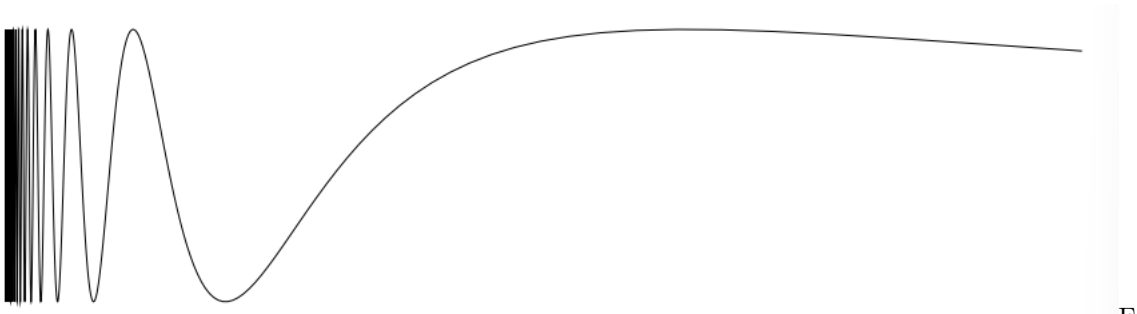


Figure 2

To show that $\overline{S}$ is not path-connected we want to show that there is no path $\gamma : [0,1] \to \bar{S}$ with $\gamma(0) \in \{0\} \times [1,1]$ and $\gamma(1) \in S$. Suppose γ is such a path, then as $\{0\} \times [-1,1]$ is closed, its pre-image is closed in $[0,1]$ so it has a largest element t, then $\gamma : [t,1] \cong [0,1] \to \bar{S}$ is a path which starts at $\{0\} \times [-1,1]$ and sends all other points into S. Write $\gamma(s) = (x(s), y(s))$ so that $x(0) = 0$ and for $s > 0$ we have $y(s) = \sin\left(\frac{1}{x(s)}\right)$. But then there is a sequence $s_n \searrow 0$ in $[0,1]$ with $y(s_n) = (-1)^n$ which contradicts the definition of a continuous function y defined on $[0,1]$.

Therefore $\bar{S}$ is connected but the following proposition implies it is connected.

Proposition. The closure of a connected space is connected.

Proof. Let $C \subseteq X$ be a connected subspace with closure $\bar{C}$ and let $\bar{C} = U \cup V$ be a decomposition into two disjoint open non-empty subsets. Then $C \cap U$ and $C \cap V$ are two disjoint open subsets of C so one must be empty as C is connected, say $C \cap V = \emptyset$ so that $C \subseteq U$, but then U, V are closed in $\bar{C}$ so they are closed in X so that means that in fact $\bar{C}$ must be contained in U as $\bar{C}$ is the intersection of all closed sets containing C and U is one of them, so V cannot be non-empty which is a contradiction.

□

3 Compactness

3.1 Basic properties of compactness

The good news is that many properties of compactness from this course were covered in the analysis course which means I have less work to do in preparing these notes!

Recall that in my analysis notes I wrote the following:

"Recall (level 6.2, 6.4) that if a subset of $\mathbb{R}^n$ is closed and bounded, then it has the following 2 properties:

1. Every sequence in the set has a subsequence converging to a point in the set

2. Every cover of the set by open sets has a finite subcover.

In general normed spaces, we call property 1 sequential compactness and property 2 compactness. We will show that in normed spaces, and in the more general notion of metric spaces, these are equivalent. You should keep in mind that they are indeed equivalent in metric spaces - we will prove it in the chapter on metric spaces. However, they are not equivalent in general topological spaces, which we will encounter in the relevant sublevel."

Well now we're at that relevant sublevel.

Definition. A topological space X is compact if every open cover has a finite subcover and sequentially compact if every sequence has a convergent subsequence.

Sequentially compact but not compact: Let X be the sets of all subsets of $\mathbb{R}$ that are at most countable. For each $A \in X$ put $U_A = \{B \in X : B \subseteq A\}$ and give X the topology with basis $\{U_A : A \in X\}$. Then this is a basis as it is closed under finite intersections. To see that this is sequentially compact, note that for a sequence in A_n, the point in X consisting of the set $\bigcup_{n=1}^{\infty} A_n$ is a limit for the sequence, but non-compactness follows as the family $\{U_A : A \in X\}$ is an open cover but given finitely many members $U_{A_1}, U_{A_2}, ..., U_{A_m}$ choose $r \in \mathbb{R} \setminus (A_1 \cup A_2 \cup ... \cup A_m)$, then the point $\{r\} \in X$ belongs to no element of this finite subcover.

To find an example of a space that is compact but not sequentially compact is harder, so we will not fully do this here. The space $\{0,1\}^{P(\mathbb{N})}$ with the product topology and P denotes the power set works, but requires the statement that an infinite product of compact spaces is compact with the product topology - we will prove this in level 11 as the proof requires some new ideas. Assuming that $\{0,1\}^{P(\mathbb{N})}$ is indeed compact, we can show that it is not sequentially compact. To do this, note that our space is of functions $P(\mathbb{N}) \to \{0,1\}$, so now consider the sequence of functions $f_n(A)$ that is 1 if $n \in A$ and 0 if $n \notin A$ for any $A \subseteq \mathbb{N}$. Then this has no convergent subsequence because if it had a sequence $f_{n_k}(x)$ then this cannot be eventually constant in every coordinate (needed for convergence by properties of the product topology) by considering the set that is 1 at $n_{k_{\text{odd}}}$ and 0 at $n_{k_{\text{even}}}$ and does whatever for all other natural numbers.

Example. Any space with the cofinite topology is compact as given one thing in an open cover, we can get a finite subcover by for each missing point in that first thing picking something else in the cover that covers that missing point.

Proposition. If X is a compact topological space and $C \subseteq X$ is a closed subspace then C is compact.

Proof. If $\{U_a : a \in I\}$ is an open cover of C then adding $X \setminus C$ gives an open cover of X that has a finite subcover by

compactness and removing $X \setminus C$ from that finite subcover if necessary gives a finite subcover of C.

□

Proposition. If X is a Hausdorff topological space and $C \subseteq X$ is a compact subspace then C is closed.

Proof. We will show that $X \setminus C$ is open. Fix a $x \in X \setminus C$, then for each $y \in C$ there exist disjoint open sets V_y containing x and W_y containing y by Hausdorffness. Then the collection $\{C \cap W_y : y \in C\}$ is an open cover of C so by compactness it has a finite subcover. Let $y_1, y_2, ..., y_n \in C$ be such that $\{C \cap W_{y_1}, C \cap W_{y_2}, ..., C \cap W_{y_n}\}$ is a finite subcover, then $\bigcap_{i=1}^{n} V_{y_i}$ is a finite intersection of open sets containing x, so open, but disjoint from all W_{y_i}, thus disjoint from C, so for each $x \in X \setminus C$ there is an open neighbourhood around it disjoint from C which means $X \setminus C$ is open so C is closed.

□

By the above 2 propositions, in compact hausdorff spaces the closed subsets are exactly the compact subsets.

Proposition. If X is a compact topological space and $f : X \to Y$ is a continuous function then $f(X)$ is a compact subspace of Y.

Proof. Assume without loss of generality that $Y = f(X)$ and that f is surjective. Let $\mathcal{U}$ be an open cover of $Y = f(X)$ and set $f^{-1}(\mathcal{U}) = \left\{f^{-1}(U) : U \in \mathcal{U}\right\}$, then this is an open cover of X which has a finite subcover by compactness of X, so suppose that $f^{-1}(U_1), f^{-1}(U_2), ..., f^{-1}(U_n)$ cover X. Then $U_i = f\left(f^{-1}(U_i)\right)$ cover $f(X) = Y$ and is a finite subcover of $\mathcal{U}$ so $f(X)$ is compact.

□

Proposition. If f is a continuous bijection from a compact topological space X to a Hausdorff topological space Y then f is a homeomorphism.

Proof. We want to show that $f^{-1} : Y \to X$ is continuous and we therefore want to show that $\left(f^{-1}\right)^{-1}(C)$ is closed in Y whenever C is closed in X. Luckily we don't have to think about pre-images since f is a bijection so we just have $\left(f^{-1}\right)^{-1}(C) = f(C)$. If $C \subseteq X$ is closed then it is compact by the first proposition above and thus $f(C) \subseteq Y$ is compact by the third proposition above and thus closed by the second proposition above and the fact that Y is Hausdorff.

□

Proposition. (Extreme value theorem) Let X be a compact topological space and let $f : X \to \mathbb{R}$ be continuous, then there exist $a, b \in X$ such that $f(a) \leq f(x) \leq f(b)$ for all $x \in X$.

Proof. The set $f(X) \subseteq \mathbb{R}$ is compact by an earlier proposition so it is closed and bounded and attains its bounds by closedness.

□

3.2 Compactness of products

Although we will not prove here that compactness holds for arbitrary products, we will prove that it holds for finite products.

Lemma. Let X and Y be compact topological spaces and $x_0 \in X$ and suppose that W is an open set containing $\{x_0\} \times Y$. Then there is an open neighbourhood U_{x_0} containing x_0 in X such that $U_{x_0} \times Y \subseteq W$.

Proof. We can cover $\{x_0\} \times Y$ by its intersection with open sets of the form $U \times V \subseteq W$ for U open in X and V open in Y because such sets are a basis for the product topology on $X \times Y$. Compactness of $\{x_0\} \times Y$ implies that this has a finite subcover so pick $U_1 \times V_1, U_2 \times V_2, ..., U_n \times V_n$ inside W that still covers $\{x_0\} \times Y$. Then $U_1 \cap U_2 \cap ... \cap U_n$ is the desired U_{x_0}.

Figure 3 shows how and why this proof works after taking the finite subcover.

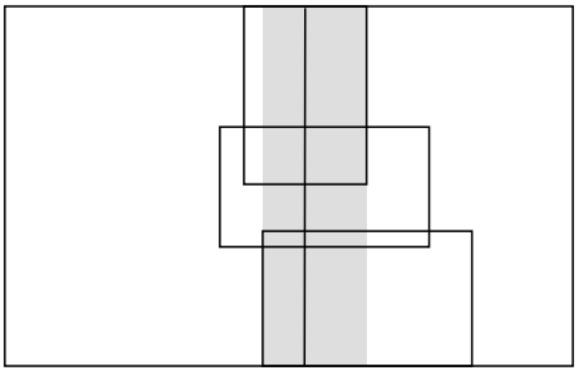


Figure 3

□

Theorem. Finite products of compact sets are compact.

Proof. We will show that X, Y compact implies $X \times Y$ compact as by induction this extends to finite products.

Let W be an open cover of $X \times Y$. For fixed $x_0 \in X$ we have $\{x_0\} \times Y$ is compact so it may be covered by its intersection with finitely many $W_1, W_2, ..., W_n \in W$ which by the above proof contains some $U_{x_0} \times Y$. Do this for each x and then by compactness, X has a finite cover of these U_x's which gives a finite collection of these finite covers that covers $X \times Y$.

□

Note that $[0, 1]$ can be shown to be compact directly, and by this theorem it follows that $[-N, N]^d$ is compact, so by above discussion we can deduce (in a topological way, so different from the way we did in analysis) that compact subsets of $\mathbb{R}^n$ are exactly those which are closed and bounded.

3.3 The finite intersection property

3.3.1 Basic properties

Definition. A collection $\{S_\alpha : \alpha \in I\}$ of subsets of some set is said to have the finite intersection property if every finite intersection of the S_α's is non-empty.

Theorem. A topological space X is compact if and only if every collection $\{C_a : a \in I\}$ of closed sets with the finite intersection property has $\bigcap_{a \in I} C_a \neq \emptyset$.

Proof. Suppose that X is compact and let $\mathcal{U} = \{X \setminus C_a : a \in I\}$ be a collection of open subsets. If we suppose that $\bigcap_{a \in I} C_a = \emptyset$ then $\mathcal{U}$ is an open cover, as every point of X lies outside some C_a. By compactness it has a finite subcover which means that for this subcover $\bigcap_{i=1}^{n} C_{a_i} = \emptyset$. This means that if X is compact and $\bigcap_{a \in I} C_a \neq \emptyset$ does not hold for some collection of closed sets then that collection does not have the finite intersection property i.e. X compact implies every collection $\{C_a : a \in I\}$ of closed sets with the finite intersection property has $\bigcap_{a \in I} C_a \neq \emptyset$.

On the other hand, if X has that every collection of closed sets with the finite intersection property has $\bigcap_{a \in I} C_a \neq \emptyset$ and $\mathcal{U} = \{U_a : a \in I\}$ is an open cover then $\{X \setminus U_a : a \in I\}$ is a collection of closed sets with empty intersection so it cannot have the finite intersection property, so there is some finite list $X \setminus U_{a_i} : i = 1, 2, ..., n$ with empty intersection giving a finite subcover so X is compact.

□

3.3.2 Discussion and properties of the cantor set

Now recall the cantor set - We have briefly discussed this in earlier analysis levels. As a reminder, it's the set constructed as in figure 4, equivalently the elements of $[0,1]$ with no 1's in their base 3 expansion.

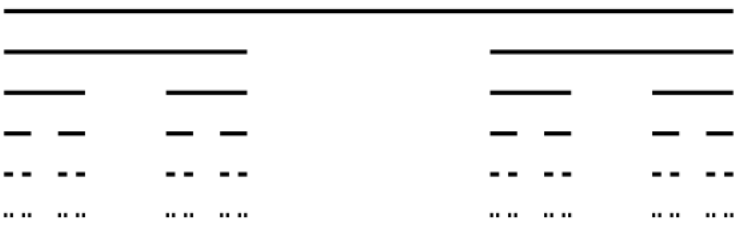


Figure 4

This set is known for having measure 0 (exercise) and being uncountable (because there is a natural bijection between it and the power set of $\mathbb{N}$ given as after you fix a subset of $\mathbb{N}$ writing $0.x_1x_2...$ in base 3 where x_i is twice the indicator function of i in the subset of $\mathbb{N}$ in question).

Note that the cantor set is closed as the intersection of closed sets from figure 4. Also note that the endpoints are NOT the only elements, since $\frac{1}{4}$ is in there and it is reached by going "left half right half left half right half ... " and this point corresponds to the set of all even numbers.

I love the cantor set! The reason is that it has so many weird properties and is a counterexample to a lot of things, and in particular this means that if the cantor set comes up we can be pretty confident that we are definitely not doing applied maths.

Definition. A topological space is said to be totally disconnected if its connected components all consist of a single point.

Proposition. The cantor set is totally disconnected.

Proof. Let C denote the cantor set. If $x, y \in C$ are distinct points with $x < y$ then there is a point $z \in [x, y]$ that is not in C. This is kind of obvious, as after $\log_3\left(\frac{1}{y-x}\right)$ steps there will be no interval of length $y - x$ left in the set.

□

Also, if we divide by 4 instead of 3 before removing points, we get the "fat cantor set" which no longer has measure 0 but yet is homeomorphic to the cantor set. This has applications in analysis - see level 7 bonus where we use this to construct a non-integrable derivative.

4 Manifolds

4.1 Basic properties

Definition. A topological manifold of dimension d is a topological space X which is Hausdorff and such that every point of X has an open neighbourhood which is homeomorphic to (an open subset of) $\mathbb{R}^d$.

Note that this implies every point has a neighbourhood homeomorphic to an open ball in $\mathbb{R}^d$ by taking a sub-neighbourhood, and this is homeomorphic to $\mathbb{R}^d$ itself. We can therefore alternatively write

Definition. A topological manifold of dimension d is a topological space X which is Hausdorff and such that every point of X has an open neighbourhood which is homeomorphic to $\mathbb{R}^d$.

Example. An open subset of $\mathbb{R}^d$ itself is a manifold.

Example. The sphere $S^d \subseteq \mathbb{R}^{d+1}$ is a topological manifold of dimension d. To see that it is locally euclidean, note that it is spherically symmetric so we will just do it about $p = (-1, 0, 0, ..., 0)$. Then the local homeomorphism from $S^d : x < 0 \to B_1(0)$ by simply projecting the point onto $\mathbb{R}^d$.

Example. The circle S^1 is given by taking $[0, 1]$ and declaring $0 \sim 1$ and taking the quotient.

Example. The "line with two origins" is locally homeomorphic to $\mathbb{R}$ but it is not a manifold because it is not Hausdorff.

Example. If M, N are manifolds then $M \times N$ is also a manifold as if $(m, n) \in M \times N$ then if $m \in U \subseteq M$ and $n \in V \subseteq N$ where U, V are homeomorphic to $\mathbb{R}^m$ and $\mathbb{R}^n$ respectively then (m, n) has a neighbourhood $U \times V$ homeomorphic to $\mathbb{R}^{m+n}$. It is an exercise to check that products of Hausdorff spaces are Hausdorff.

Example. Consider $T = [0, 1] \times [0, 1] / \sim$ where $\sim$ is given by $(0, t) \sim (1, t)$ and $(s, 0) \sim s(1)$ so that it looks like figure 5.

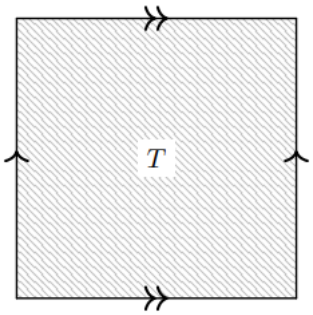


Figure 5

Then by gluing the ends together twice we get a donut!

Example. Consider the equivalence relation on $[0, 1] \times \mathbb{R}$ given by $(1, v) \sim (0, -v)$ with quotient space M. Then this is the mobius band and you can see it in Figure 6.

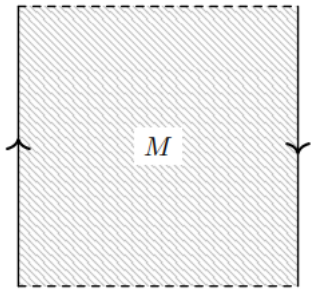


Figure 6

This is a 2-dimensional manifold.

The quotient space K of $[0, 1]^2$ shown in Figure 7 with the arrows depicting the equivalence relation is called the Klein bottle.

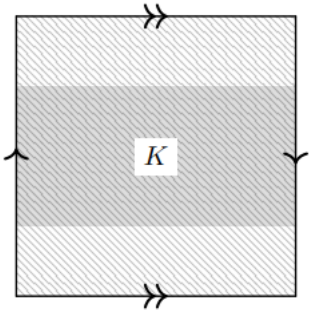


Figure 7

Theorem. A topological manifold is connected if and only if it is path connected

Proof. We know that path connected implies connected in general so suppose that X is a connected topological manifold. The path components of X partition it into disjoint subsets, so to show that there is only one path component it suffices to show that the path components are open. To do this we will show that for each $p \in X$ there is an open neighbourhood of p consisting of points which can be connected to p by a path, but wait, this follows from the fact that X locally looks like $\mathbb{R}^d$, so done.

□

There is another result we will prove about manifolds but we need a lemma first.

Lemma. If $p, q \in B_1(0) \subseteq \mathbb{R}^d$ then there is a homeomorphism ϕ of $\mathbb{R}^d$ sending $p \to q$ which is the identity outside of $B_1(0)$.

Proof. Assume $p = 0$ because if there is such a homeomorphism ϕ sending $0 \to q$ and another one ψ sending $0 \to p$ then $\phi \circ \psi^{-1}$ sends p to q. Consider the map $\rho : x \mapsto \frac{x}{1+|x|}$ mapping $\mathbb{R}^d$ into $B_1(0)$ with inverse $\frac{y}{1-|y|}$. Then there is a homeomorphism $x \mapsto x + \rho^{-1}(q)$ of $\mathbb{R}^d$ and when conjugated with ρ this gives a homeomorphism $B_1(0) \to B_1(0)$ sending $0 \to q$.

From here, the details are annoying, but essentially the idea is that if $\mathbb{R}^2$ (for example) is a grid then it looks like figure 8, and if $q = \frac{1}{2}$ it transforms into figure 9 as the red gridlines move half a gridstep forward, and we see that therefore this homeomorphism $B_1(0) \to B_1(0)$ sending $0 \to q$ extends to a homeomorphism ϕ of $\mathbb{R}^d$.

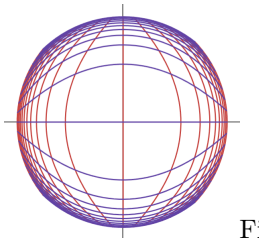


Figure 8

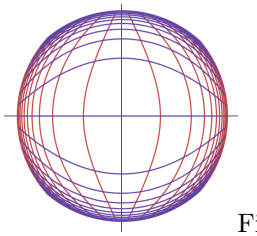


Figure 9

□

Using this lemma we can prove:

Theorem. Let X be a connected topological manifold. Then for any two points $p, q \in X$ there is a homeomorphism $\phi : X \to X$ such that $\phi(p) = q$.

Note that it is connected as the disjoint union $S^1 \sqcup \mathbb{R}$ certainly fails this as you can't map from the circle into the line in a homeomorphic way.

Proof. Let $p \sim q$ if there is a homeomorphism ϕ sending $p \to q$. Then this is an equivalence relation sending X into disjoint subsets, so by connectedness it is sufficient to show that these equivalence classes are open. We will do this by showing that for each $p \in X$ there is an open neighbourhood of p consisting of points which can be reached from p by applying a Homeomorphism.

It is tempting to say this is true by the manifold property as in a local copy of $\mathbb{R}^d$ about p, the subcopy of $B_1(0)$ is such a neighbourhood by the lemma above, but we must check that this gives a globally continuous map. To do this, we will explain this idea more precisely and also add an application of the gluing lemma.

Let $p \in U$ and let U be homeomorphic to $\mathbb{R}^d$ by a homeomorpism $\tau : U \to \mathbb{R}^d$ sending p to 0, then set $C := \tau^{-1}\overline{B_1(0)} \subset U$. Then for each q in the interior of C which is exactly $\tau^{-1}B_1(0)$, the lemma shows there is a homeomorphism ϕ_q of $\mathbb{R}^d$ sending 0 to $\tau(q)$ which is the identity outside $B_1(0)$. Then $\tau^{-1} \circ \phi_q \circ \tau$ is a homeomorphism of U sending $p \to q$ which is the identity outside C (which is well defined as $\tau^{-1} \circ \phi_q \circ \tau$ is identity outside C anyway). Define a function ψ_q to be this in U and identity everywhere else, then we want to show that ψ_q is a Homeomorphism. This map we know is continuous restricted to U and restricted to $X \setminus C$, and C is compact and X is Hausdorff thus C is closed, so by the gluing lemma ψ_q is continuous and similarly ψ_q^{-1} is continuous.

□

So far, the examples of manifolds we have seen can be realized as subspaces of some $\mathbb{R}^N$. It turns out that this is always true whenever the manifold is compact.

Theorem. Let X be a compact d-dimensional topological manifold. Then there is a $N > 0$ and a subspace $Y \subseteq \mathbb{R}^N$ homeomorphic to X.

Proof. Let $\sigma : \mathbb{R}^d \to [0, \infty)$ be continuous and 1 on the closed unit ball and 0 outside $B_2(0)$. It is easy to see that such a function exists.

For each $x \in X$ there is a neighbourhood $x \in U_x$ and a homeomorphism $\tau_x : U_x \to \mathbb{R}^d$. Let the open set $V_x \subseteq U_x$ correspond under τ_x to $B_1(0) \subseteq \mathbb{R}^d$ and let $\sigma_x = \sigma \circ \tau_x : U_x \to [0, \infty)$. There is a function $b_x : X \to \mathbb{R}^d$ defined by

$$y \mapsto \begin{cases} \sigma_x(y)\,\tau_x(y) & y \in U_x \\ 0 & y \notin U_x \end{cases}$$

On the open set $X \setminus \tau_x^{-1}\overline{B_2(0)}$ (this is open as compact sets are closed in a Hausdorff space) this function is identically 0 so is continuous, and it is continuous on U_x so it is continuous by the gluing lemma. Similarly we may consider σ_x as a continuous function $X \to [0, \infty)$ by making it 0 outside U_x. The collection $\{V_x : x \in X\}$ is an open cover of X so there is a finite subcover $V_{x_1}, V_{x_2}, ..., V_{x_n}$. Now consider the function

$$e : X \to \left(\mathbb{R}^d \times \mathbb{R}\right)^n$$

defined by

$$y \mapsto \left(b_{x_1}(y), \sigma_{x_1}(y); b_{x_2}(y), \sigma_{x_2}(y); ...; b_{x_n}(y), \sigma_{x_n}(y)\right)$$

This is continuous as each component is continuous. If $e(y) = e(y')$ then suppose $y \in V_{x_i}$ since the V's cover so that $\sigma_{x_i}(y) = 1$ and hence $\sigma_{x_i}(y') = 1$ too. Therefore,

$$\tau_{x_i}(y) = b_{x_i}(y) = b_{x_i}(y') = \tau_{x_i}(y')$$

since $\beta = \tau$ when $\sigma = 1$. This means $y = y'$ as τ is a homeomorphism, thus e is injective. Writing $Y := e(X)$, we have that $e : X \to Y$ is continuous. It is a bijection from a compact to a Hausdorff space so it is a homeomorphism.

□

Note that quotients of compact spaces are compact, so anything I defined by an equivalence relation on $[0,1]^2$ does embed into Euclidean space by the above theorem. But the theorem is conservative in the sense that S^2 clearly is in $\mathbb{R}^3$ but the best you can do with the above theorem is $\mathbb{R}^4$. Getting the minimal dimension is beyond the scope of this course.

As one more example, consider the real projective space which is $\mathbb{R}^n \setminus 0$ defined by $x \sim y$ if $x = \lambda y$ for any λ, not just $\lambda > 0$. This is $\mathbb{RP}^{n-1}$. Specializing to $n = 3$, this is nothing but the points on the sphere quotiented by $x \sim -x$. This is actually just the upper hemisphere with these antipodal points identified, so we can draw it as another diagram as in figure 10.

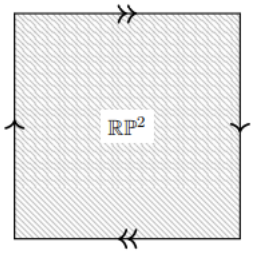


Figure 10

4.2 Topological surfaces and triangulation

This will be a very brief introduction to triangulation - we will do a lot more of this in algebraic topology.

Definition. A topological surface is a 2-dimensional topological manifold. We have seen some examples of this already.

Note that we can obtain S^2 with the following equivalence relation. To see why, get a square paper and fold the edges together so the arrows match in figure 11 and you will see it is like the sphere.

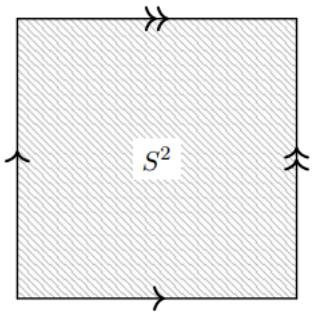


Figure 11

In general we can take a polygon and glue edges together to get a compact topological surface.

A triangle in $\mathbb{R}^N$ is the smallest convex set containing 3 points (their "convex hull"). A finite collection of triangles is said to form a polyhedron if each intersection $\triangle_i \cap \triangle_j$ with $i \neq j$ is either empty, a common vertex of both triangles, or a common edge of both triangles, figure 12 shows some examples and non-examples.

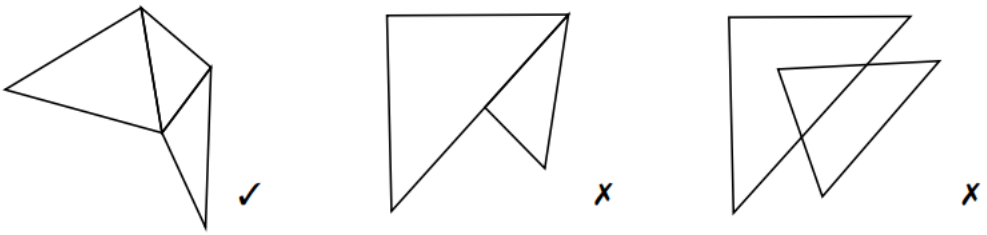


Figure 12

Definition. A triangulation of a topological surface S is a polyhedron K and a homeomorphism $\phi : |K| \to S$.

Example. To triangulate the torus, we consider Figure 13 works. The reason we need so many triangles is because we need it to satisfy the equivalence relation while it is also the case that each triangle in the picture is uniquely determined by its vertices.

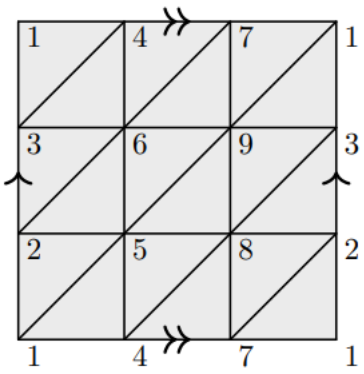


Figure 13

We cannot do this with a 2×2 grid because then the bottom right and top left triangles would have the same vertices and therefore they would not be disjoint in a euclidean embedding (meaning "subspace of $\mathbb{R}^n$ corresponding to") of this.

It is possible to check that this is indeed a polyhedron. This gives a homeomorphism to a polyhedron in $\mathbb{R}^9$ by assigning each vertex to a basis vector and then making the corresponding triangles. This puts the torus in $\mathbb{R}^9$ which we know is conservative as it is in fact in $\mathbb{R}^3$. See figure 14 for a triangulated surface in $\mathbb{R}^3$.

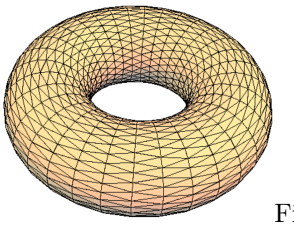


Figure 14